

Ultrametric Quantum Error Correction

Testing Discrete Geometry in the Redundancy–Fidelity Tradeoff · CWI Summer School on QA & QEC · 24–28 August 2026 · Poster
Session Wed 26 Aug

Core Idea

Standard QEC assumes noise accumulates over a continuous topology and fidelity scales smoothly with redundancy. We are testing an alternative: error propagation follows a discrete, hierarchical Bruhat–Tits tree geometry, where redundancy quantizes in units of $(p+1)$. **Signature:** a "staircase" of flat plateaus and sudden fidelity jumps instead of a smooth curve. *Hypothesis only — no result claimed.*

Proposed Protocol (not yet run)

1. Encode one logical qubit in the Shor $[[9,1,3]]$ code.
2. Interact with a structured spectator-qubit environment.
3. Correct and measure final logical fidelity.
4. Group environment qubits into larger hierarchical fragments; track fidelity across fragment boundaries.

The Cross-Check Already Run — and It Mostly Failed (reported in full)

A 2-adic valuation witness ($v_p^{\max}$ of the Mahler spectrum) separates the Golay CSS $[[23,1,7]]$ code ($v_p^{\max} = 28$) from random codes (~ 4) — but does **NOT** generalize: Shor (2), Steane (2), Perfect (1), surface codes (2–6) all cluster at the random baseline. The gap is a Golay-symmetry artifact, not an optimality witness. **Open:** which witness can detect ultrametric structure?

The Diagrammatic Bridge (ZX-calculus) — same map, the room's notation

Spiders (the atom): green = Z-basis copy/merge, red = X-basis. Graph-state convention: green spider (0 in, 1 out) = $|+\rangle$; yellow H-box (2 in, 2 out) = CZ = the graph-state edge. Figure: the Bruhat–Tits tree T_2 as a ZX graph state (22 spiders = 22 vertices, 21 H-boxes = 21 edges).

Pauli webs (the stabilizers, read off the diagram): a connected sub-diagram that evaluates to a Pauli operator — the code's stabilizers and logical operators as paths through the spider web (Wan–Price–Yao, arXiv:2601.04467).

Gadgets (the field's cost unit): phase/Pauli gadgets $e^{i\alpha P}$; T-count optimization is gadgetization (arXiv:1902.03178; Pauli Gadget Synthesis, QPL 2026). The algebraic cousin of an energy cost — the seam this poster cares about.

The one-line statement: "I represent the ultrametric geometry as a ZX graph state — the same language as the holographic-code construction on tree-like tessellations (Wan–Price–Yao 2026). The stabilizers are the Pauli webs of the diagram; the hierarchy is the tree."

Honest caption (map $\neq$ territory): the diagrams import structure from different silos — internally consistent, externally unprobed. This handout says so. *"It's a map, not the territory — but it's the same map the field draws."*

Working Papers & References

Zenodo concept DOIs: Ultrametric Code Spaces — 10.5281/zenodo.21824194 · Archimedean Shadows (QEC–Darwinism) — 10.5281/zenodo.21809888 · Number-Theoretic Ultrametric Foundations (C7.3) — 10.5281/zenodo.21193003 · ACRP-06 $v_p^{\max}$ extension — 10.5281/zenodo.21737221 · Adelic Cross-Domain Program v5 — 10.5281/zenodo.21691414 · "What Remains" (computing/QEC) — 10.5281/zenodo.21922812

ZX references (arXiv, verified 2026-08-18): Holographic codes seen through ZX-calculus — arXiv:2601.04467 · Iteratively decoded magic state distillation — arXiv:2410.17992 · ZX-calculus for the working quantum computer scientist — arXiv:2012.13966 · Graph-theoretic simplification with ZX-calculus — arXiv:1902.03178

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